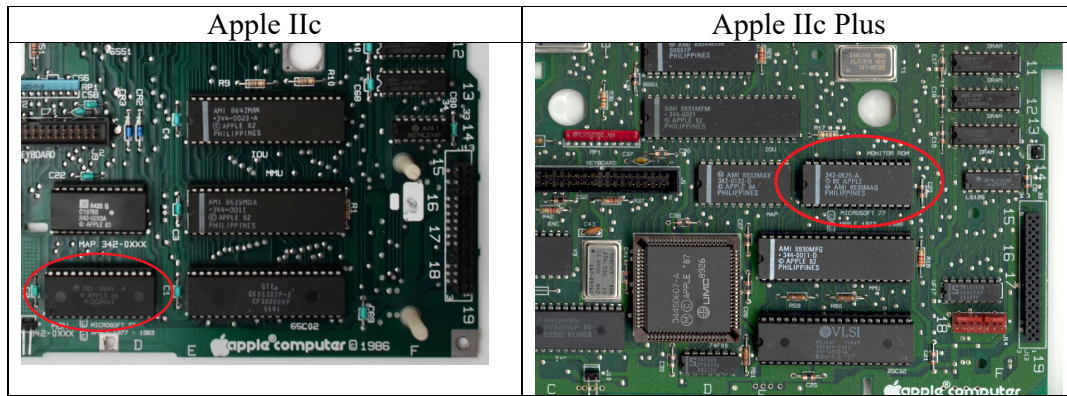


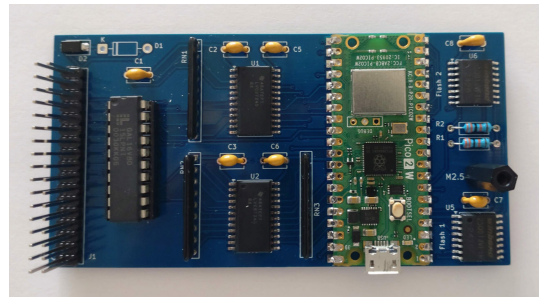
MegaFlash User Guide

Installation

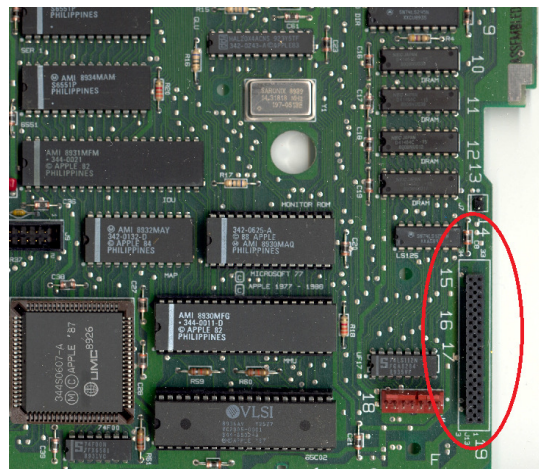
1. Replace the Monitor ROM with the supplied ROM chip.



2. For Rev 1.1 PCB, it is recommended that an 18mm nylon standoff is installed to the mounting hole labeled M2.5 to support the card as shown below.



3. Install MegaFlash to the Memory Expansion Connector of Apple IIc/IIc plus motherboard. Make sure the header pins are aligned to the connector correctly.



Boot Menu and Control Panel

- Ctrl-Option-Reset to enter the boot menu.
- In the boot menu, there are some shortcut keys:
 - C key to enter control panel
 - Space bar to toggle CPU speed on Ilc+
 - Esc key to reboot

Booting from drive 2

IN#4 command to boot from drive 2 of MegaFlash

Control Panel

Drives Enable

Each flash drive and the RAM Disk can be individually enable/disable. Select **Drives Enable** from the main menu to open the window. Press the number keys to enable/disable the drives.

To save the changes, press Enter to return to main menu. Then, select **Save and Reboot**.

To discard any changes, press Esc key.

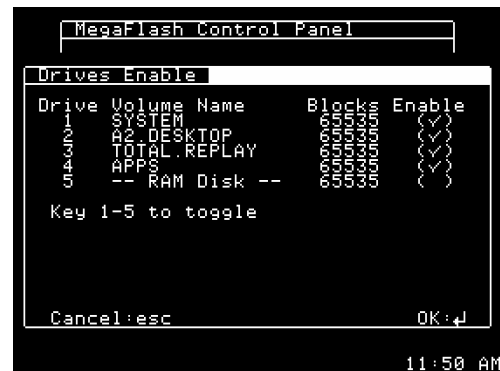
Warning: The slot/drive mappings will be changed after drives are enabled/disabled. Always identify the drives with ProDOS volume name instead of slot/drive number.

Format

It can format the flash drives and RAM drive on MegaFlash. It has two special features. 1) It can format drives to any arbitrary size. 2) It can erase the entire drive before formatting. It may take up to 2 minutes to erase a flash drive. Subsequent write access will be much faster if the drive is already erased.

MegaFlash Hardware/Firmware Information

To show the hardware and firmware information, go to the boot menu. Hold the Open-Apple key while entering Control Panel. i.e. OA-7 or OA-C. Hold Open-Apple key until the device information screen appears.



```
Device Information
=====
Pico Board = Pico 2 RP2350
Wifi Supported = Yes
MegaFlash Pico Firmware Version = U1.1.1
Firmware Build Timestamp = Aug 4 2025 16:48:32
Pico SDK Version = 2.2.0
Total Flash Capacity = 256MB
Flash Chip #0 JEDEC ID = EF7021h
Flash Chip #1 JEDEC ID = EF7021h
Control Panel Build Timestamp = Aug 4 2025 16:45:44
Press any key to continue
```

Apple IIc+ Accelerator

The accelerator code has been modified:

- Pressing Esc key on Power up to switch to normal mode is *removed*.
- To change CPU speed, first go to Boot Menu. Press key 8 or Space bar to toggle the CPU speed. Then, reboot
- Ctrl-OA-Reset *does not* change the current CPU speed.
- The word **Normal** is shown when CPU is running at 1MHz during cold boot.
- The speed when the computer is powered up can be set in Control Panel

ROM Disk

- A ROM Disk is provided for disaster recovery. The ROM disk contains Copy II+ and ADT Pro.
- When enabled from boot menu, it will become the first unit of MegaFlash (slot 4, drive 1). Please note that the slot/drive assignments of all MegaFlash units are changed.
- Ctrl-OA-Reset cold boot to disable the ROM Disk

RAM Disk

- RAM Disk can be enabled from Control Panel. The size of RAM Disk is 400kB. RAM Disk is the last unit of SmartPort. Its volume name is /RAMDISK.
- RAM Disk can be used to transfer floppy disk image with ADT Pro or TFTP. Users may transfer a floppy disk image and write it to the RAM Disk. Then, copy files from the disk image to Flash storage.

Applesoft BASIC FPU

- High bit of address \$5FC is the Enable flag of Applesoft FPU.

Real Time Clock and Network Time Sync

The Raspberry Pi Pico has a real time clock (RTC). But it does not support battery backup. MegaFlash gets the time from NTP server via WIFI.

To enable network time sync,

- 1) Configure the **WIFI Settings** from Control Panel
- 2) Run **Test Wifi/NTP** from Control Panel
- 3) If all tests are passed, enable the **Network Time Sync** option and setup the time zone.

If MegaFlash can fetch the time from NTP server, the time is shown at the bottom right corner in Boot Menu and Control Panel.

If Network Time Sync option is enabled, a ProDOS clock driver will be installed automatically. The clock driver is in the patched system ROM. No software installation is required.

If you want to install your own clock device driver, please disable Network Time Sync.

Slinky Emulation

With the stock Apple IIc firmware, MegaFlash can emulate a 256kB Memory Expansion card (aka Slinky) if Pico 2 controller board is used.

Disk Image Transfer

Disk Image File can be transferred to MegaFlash in 3 ways.

- 1) ADT Pro via IIC RS-232 Serial Port
- 2) MegaFlash Terminal Mode via USB on Pico
- 3) TFTP Server via WIFI

Terminal Mode

- To enter terminal mode, TURN OFF Apple IIc. Then, connect a PC to USB port of Pi Pico. Connect to MegaFlash serial port with a Terminal Program like TeraTerm (Windows).
- In the terminal mode, ProDOS-order disk image (`.po` or `.hdv`) can be uploaded to or downloaded from MegaFlash using XMODE-CRC or XMODEM/1K protocol.

```
COMS - Tera Term VT
File Edit Setup Control Window Help
3) Download ProDOS Image file from MegaFlash
4) Erase Flash Content

Please Select:
```



```
#          #              #####             #          #          #          #          #
##        ##      #####   ###            #           #           #       ###    #      #
#         #      #####   #               #           #           #       #       #      #
#        #      #####   #               #           #           #       #       #      #
#         #      #       #       #       #           #           #       #       #      #
#         #      #       #       #       #           #           #       #       #      #
#         #      #####   #               #           #           #       #       #      #
#         #      #####   #               #           #           #       #       #      #
```



```
Main Menu
=====

1) Device Information
2) Upload ProDOS Image file to MegaFlash
3) Download ProDOS Image file from MegaFlash
4) Erase Flash Content

Please Select:[]
```

Disk Image Transfer via WIFI

Disk Image can be transferred between MegaFlash and TFTP server via WIFI. Note that TFTP is not FTP. TFTP protocol is much simpler and without user authentication.

To transfer disk image, setup the WIFI network from Control Panel. Then, **select Disk Image Transfer via WIFI** from Control Panel and follow the instruction.

Please note that the TFTP server and MegaFlash must be on the same subnet and TFTP traffic cannot pass through firewall or NAT. In other words, MegaFlash and the server must be connected to the same WIFI router.

The block size of original TFTP protocol is 512 bytes. Later TFTP option extension allows larger block size. A larger block size increase transfer speed dramatically. Make sure **blocksize option negotiation** is enabled when configuring the TFTP server.

From Pico firmware Version 1.1.9, TFTP Download supports DOS order disk image. If the filename extension is `.dsk` or `.do`, the image is recognized as DOS order. The sector order will be converted to ProDOS order before writing to Flash or RAM Drive.

On Windows, Tfpd64 (<https://pjo2.github.io/tftpd64/>) server program is recommended. It has a GUI. You may need to run the app as administrator to save the setting. When sending a file to server, the file is overwritten without any warning if the file already exists on the server. Make sure **Option negotiation** is enabled.

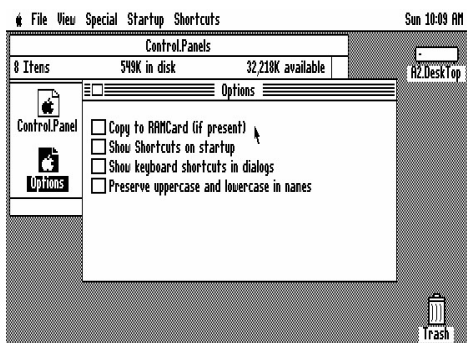
I don't have a Mac. So, I cannot test any Mac TFTP servers. But this software looks promising.

Pumpkin

<https://github.com/mterror/pumpkin>

Apple II Desktop

By default, Apple II Desktop copies itself to RAM Disk during start-up. This is not necessary for MegaFlash. This behaviour can be disabled from Apple II Desktop Control Panel. Select **Control Panel** from Apple Menu. Open **Options**, then disable **Copy to RAMCard**.



Updating Pi Pico Board Firmware

The first step is to identify the model of Pico Board on your MegaFlash. Currently, Pico W and Pico 2W boards are supported. The firmware of the two boards is different. Please prepare the correct firmware file (.uf2) for your Pico board.

To upgrade the firmware,

- 1) Remove MegaFlash from your Apple IIc computer.
- 2) Press and **hold** the white button labeled BOOTSEL on the Pico board
- 3) Connect the Pico board to a USB port of a computer while holding the button. Wait a few seconds, then release the white button
- 4) A disk drive called RP2350 or RPI-RP2 should be mounted.
- 5) Drag-and-drop the firmware file to the drive. i.e. Copy the .uf2 file to the root directory of the drive. Wait until the file copy finishes.
- 6) Reinstall MegaFlash back to your Apple IIc computer carefully.

